

$$a=2x, b=3y \quad a^3 - b^3 =$$

$$\begin{aligned} (2x)^3 - (3y)^3 &= (2x-3y) \cdot ((2x)^2 + 2x \cdot 3y + (3y)^2) = \\ &= (2x-3y) \cdot (4x^2 + 6xy + 9y^2) \end{aligned}$$

$$(a \cdot b)^n = a^n \cdot b^n$$

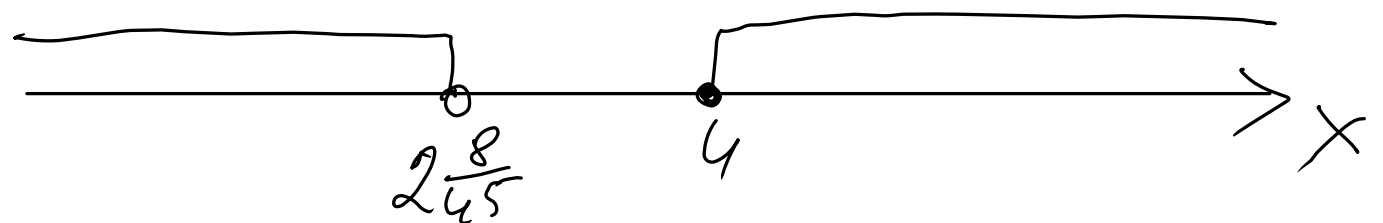
$$\begin{cases} 4(x^2 - 4x + 4) - 4 \cdot (16 - 8x + x^2) \geq 12 + x \\ 9(4x^2 - 16x + 16) + 252 > 4 \cdot (9x^2 + 6x + 1) \end{cases}$$

$$\begin{cases} \cancel{4x^2} - 16x + 16 - 64 + \underline{32x} - \cancel{4x^2} \geq 12 + x \\ \cancel{36x^2} - 144x + 144 + 252 > \cancel{36x^2} + 36x + 4 \end{cases}$$

$$\begin{cases} 16x - 48 \geq 12 + x \\ -144x + 396 > 36x + 4 \end{cases}$$

$$\begin{cases} 15x \geq 60 \quad | : 15 \\ -180x > -392 \quad | : (-180) \end{cases}$$

$$\begin{cases} \boxed{X \geq 4} \\ X < \frac{392}{180} = 2 \frac{32}{45} = 2 \frac{8}{45} \end{cases}$$



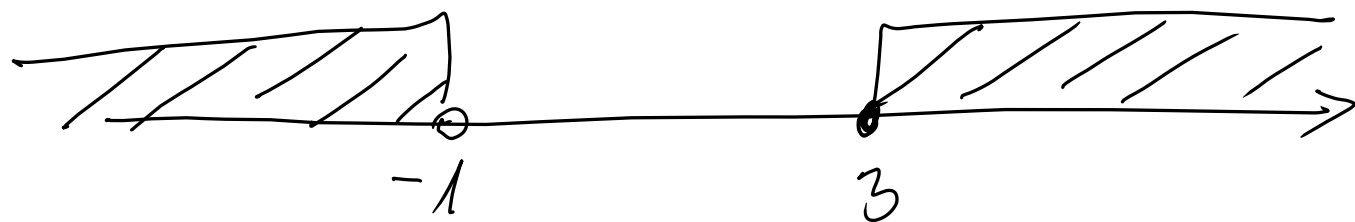
Odp: Ustod ~~to~~ nierówności nie ma rozwiązań.

Zad,

Wyznacz zbiór $A = \{x \in \mathbb{R} : x - 3 \geq 0 \vee x + 2 < 1\}$

$$x - 3 \geq 0 \quad \underline{\text{lub}} \quad x + 2 < 1 \quad \uparrow \text{lub (!)}$$

$$x \geq 3 \quad \underline{\text{lub}} \quad x < -1$$



Odp: $A = (-\infty; -1) \cup [3; \infty)$

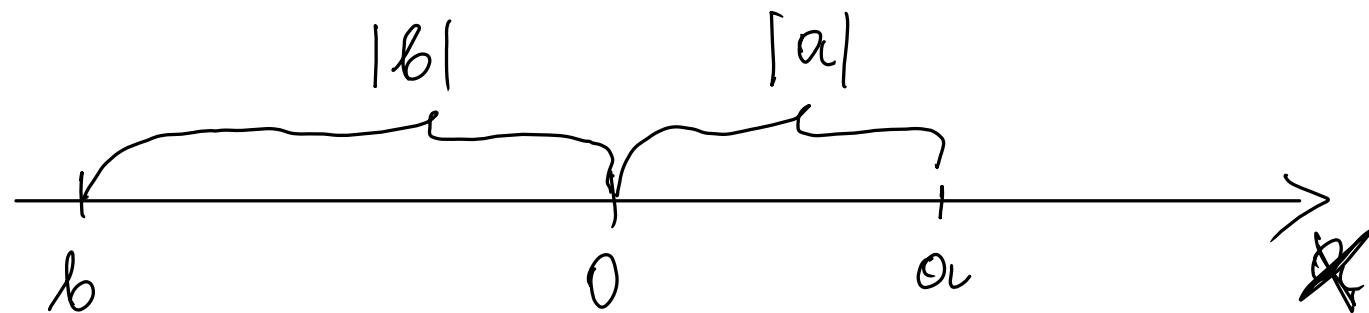
Wartość bezwzględna

$$|x| = \begin{cases} x, & \text{gdy } x \geq 0 \\ -x, & \text{gdy } x < 0 \end{cases}$$

Wniosek:

$$|x| \geq 0 \quad \text{dla } x \in \mathbb{R}$$

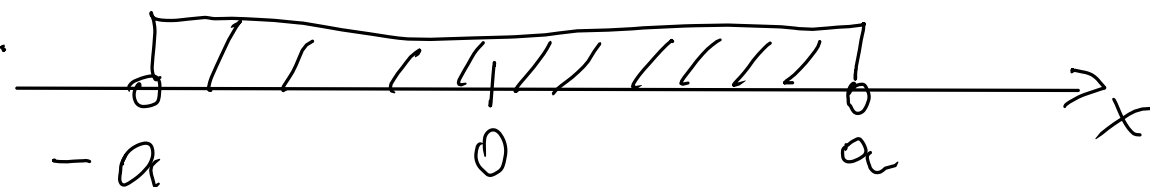
Np. $|3|=3$, $|-5|=5$, $|0|=0$



Niech $a > 0$. Wtedy

$$|x| = a \Leftrightarrow x = a \vee x = -a$$

$$|x| < a \Leftrightarrow x < a \wedge x > -a \Leftrightarrow x \in (-a; a)$$

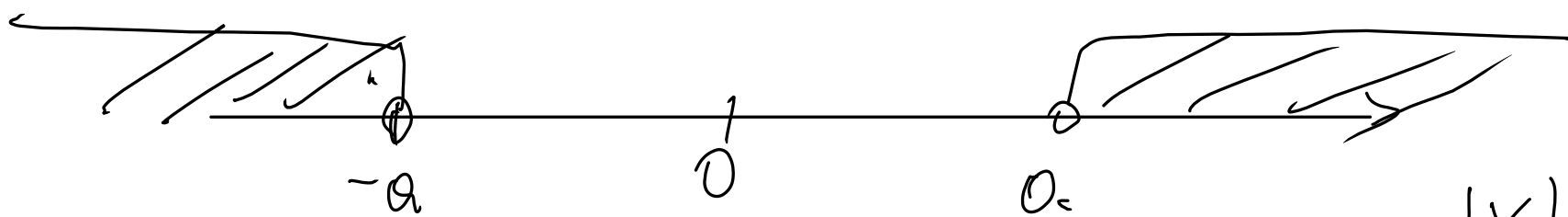


$$\underline{|x| = 4 \Leftrightarrow x = 4 \vee x = -4}$$

$$|x| < 3 \Leftrightarrow x < 3 \wedge x > -3$$

$$x \in (-3; 3)$$

$$|x| > a \Leftrightarrow x < -a \vee x > a \Leftrightarrow x \in (-\infty; -a) \cup (a; \infty)$$



$$|x| > 5 \Leftrightarrow x < -5 \vee x > 5$$

Zad. Wypisz wszystkie elementy zbioru

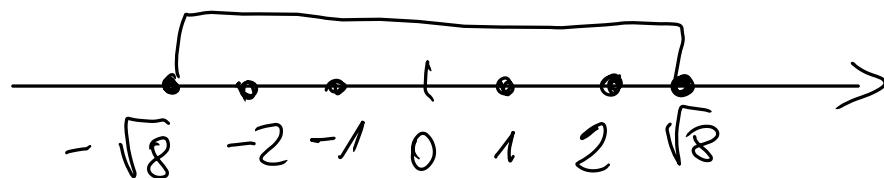
$$a) A = \{x \in \mathbb{Z} : |x| < 5\} = \{x \in \mathbb{Z} : x > -5 \wedge x < 5\} =$$

$$= \{x \in \mathbb{Z} : \underbrace{-5 < x < 5}\} = \{x \in \mathbb{Z} : x \in \underbrace{(-5; 5)}\} =$$

$$= \{-4, -3, -2, -1, 0, 1, 2, 3, 4\}$$

$$\begin{array}{ccc} 4 & \dots & 8 \cdot 9 \\ 2^2 & & 3^2 \end{array}$$

$$b) A = \{x \in \mathbb{Z} : |x| \leq \sqrt{8}\} = \{-2, -1, 0, 1, 2\}.$$



Oblicz:

$$|2\sqrt{2}+1| - 2 \cdot |\sqrt{2}-2| = 2\sqrt{2}+1 + 2 \cdot (\sqrt{2}-2) = 2\sqrt{2}+1 + 2\sqrt{2}-4 = 4\sqrt{2}-3.$$

$$|\sqrt{11}-2\sqrt{3}| \cdot (\sqrt{11}+2\sqrt{3}) = |\sqrt{11}-\sqrt{12}| \cdot (\sqrt{11}+2\sqrt{3}) = -(\sqrt{11}-2\sqrt{3})(\sqrt{11}+2\sqrt{3}) = \frac{(a-b)(a+b)}{(a-b)(a+b)} =$$

$$2\sqrt{3} = \sqrt{4} \cdot \sqrt{3} = \sqrt{12} \quad \Bigg| \quad = -[(\sqrt{11})^2 - (2\sqrt{3})^2] = -11 + 12 = 1.$$

Własności wartości bezwzględnej

$$1) \quad |x| \cdot |y| = |x \cdot y|$$

$$2) \quad \frac{|x|}{|y|} = \left| \frac{x}{y} \right|, \quad \text{dla } y \neq 0.$$

$$3) \quad |x+y| \leq |x| + |y|$$

przy czym $|x+y|=|x|+|y|$, gdy $x, y \geq 0$.

4) $\sqrt{x^2} = |x|$ Np. $\sqrt{(x+y)^2} = |x+y|$

Zad. Oblicz:

$$\sqrt{\square^2} = |\square|$$

a) $\sqrt{(1-\sqrt{3})^2} - \left| \frac{3}{2} - \sqrt{3} \right| = |1-\sqrt{3}| - \left| \frac{3}{2} - \sqrt{3} \right| =$

$$= -(1-\sqrt{3}) + \left(\frac{3}{2} - \sqrt{3} \right) = -1 + \cancel{\sqrt{3}} + \frac{3}{2} - \cancel{\sqrt{3}} = \frac{1}{2}$$

$$\left| \sqrt{64} - \sqrt{63} \right|$$

b) $\sqrt{(8-3\sqrt{7})^2} + |3\sqrt{7} - 7| = \underset{+}{|8-3\sqrt{7}|} + \underset{+}{|3\sqrt{7}-7|} =$

$$= 8 - \cancel{3\sqrt{7}} + \cancel{3\sqrt{7}} - 7 = 1.$$

Zad. Rozwiąż równanie

$$a) \sqrt{x^2 - 6x + 9} = 3$$

$$\sqrt{(x-3)^2} = 3$$

$$|x-3| = 3 \Leftrightarrow \begin{matrix} x-3=3 \\ x-3=-3 \end{matrix}$$

$$x=6 \quad \vee \quad x=0$$

$$\text{Ans: } x \in \{0, 6\}$$

$$b) \sqrt{x^2 - 8x + 16} = \frac{9}{2}$$

$$\sqrt{(x-4)^2} = \frac{9}{2}$$

$$|x-4| = \frac{9}{2} \Leftrightarrow \begin{matrix} x-4 = \frac{9}{2} \\ x-4 = -\frac{9}{2} \end{matrix}$$

$$x = 9,5$$

$$\vee \quad x = \cancel{4} - 0,5$$

$$c) \sqrt{x^2 + 10x + 25} = 1$$

$$\sqrt{(x+5)^2} = 1$$

$$|x+5| = 1 \Leftrightarrow x+5 = 1 \vee x+5 = -1$$

$$x+5 = 1 \vee x+5 = -1$$

$$x = -4 \vee x = -6$$

$$(6-x)^2 = [- (x-6)]^2 = (x-6)^2$$

$$d) \sqrt{36 - 12x + x^2} = 2$$

$$\sqrt{(6-x)^2} = 2$$

$$|6-x| = 2$$

$$6-x = 2 \vee 6-x = -2$$

$$x = 4 \vee x = 8.$$

$$|\square| = a \Leftrightarrow \square = a \vee \square = -a$$

$$e) |5x+10|=20 \Leftrightarrow 5x+10=20 \vee 5x+10=-20$$

$$5x=10 : 5 \qquad 5x=-30 : 5$$

$$\underline{x=2} \qquad \underline{x=-6}$$

$$f) \sqrt{25-10x+x^2} = 8-3|x-5|$$

$$\sqrt{(5-x)^2} = 8-3|x-5|$$

$$|5-x| = 8-3|x-5|$$

//

$$|x-5| = 8-3|x-5|$$

$$|x-5| + 3|x-5| = 8$$

$$4|x-5| = 8 : 4$$

$$|x-5| = 2$$

$$\Rightarrow \Leftrightarrow x-5=2 \vee x-5=-2$$

$$\underline{x=7} \vee \underline{x=3}$$

$$g) |4x-12| = 5 - |3-x|$$

$$4 \cdot |x-3| = 5 - |x-3|$$

$$5 \cdot |x-3| = 5 \quad | :5$$

$$|x-3| = 1$$

$$x-3 = 1 \quad \vee \quad x-3 = -1$$

$$\underline{x = 4}$$

$$\underline{x = 2}$$

Prova domo

$$1) |2x+4| \leq 6$$

$$2) |2x+3| > 1$$

